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Defining a universal and continuous Strouhal–Reynolds number relationship for the laminar vortex shedding of a circular cylinder

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The existence of a discontinuity in the Strouhal–Reynolds number relationship for the laminar vortex shedding of a cylinder is found to be caused by a change in the mode of oblique shedding. By “inducing” parallel shedding (from manipulating end conditions) the resulting Strouhal curve becomes completely continuous and agrees very well with the oblique-shedding data, if it is transformed by $S_0 = S_\theta / \cos \theta$ (where S_θ is the Strouhal number corresponding with the oblique-shedding angle θ). The curve also agrees with data from a completely different facility. This provides evidence that this Strouhal curve (S_0) is universal (for a circular cylinder).

Despite the large number of investigations on wake formation from a circular cylinder, some rather basic features of the wake remain in question, even at low Reynolds numbers when the shed vortices remain laminar. Since the first extensive measurements of vortex shedding frequencies by Roshko¹ in 1954, there has been remarkably little agreement between the many published curves of Strouhal number (S) versus Reynolds number (Re) for the laminar shedding regime (between Reynolds numbers of 49–178 in the present study). Indeed, published results have been found to differ by almost 20%. It is clear that, even though measuring the wake frequency is a simple matter, the frequency itself is very sensitive to the experimental setup, particularly to three-dimensional effects.

Some of the complexity of the problem was demonstrated by Tritton² in 1959, who found two Strouhal curves, a “low speed” mode and a “high speed” mode, separated by a discontinuity near $Re = 100$. He suggested that the discontinuity was caused by a transition between an instability originating in the wake and one originating in the immediate vicinity of the cylinder. Since that time there has been much debate over whether the discontinuity is a “fluid-mechanic” phenomenon, irrespective of the experimental setup. Berger³ discovered a discontinuity around $Re = 130$, but proposed that his discontinuity was related to the turbulence level of the wind tunnel. A further explanation was put forward by Gaster,^{4,5} who suggested that the discontinuities could be caused by a nonuniform mean flow speed across the test section, a suggestion that recently found support from Mathis *et al.*⁶ Gerrard⁷ suggested that his Strouhal discontinuity at $Re = 100$ was, in some way, related to the end of a regime of Re in which diffusion of vorticity plays a primary role in the vortex shedding. More recently, Sreenivasan⁸ interpreted several of his Strouhal discontinuities as being involved in the “route to chaos” in the cylinder wake. Subsequently, Van Atta and Gharib⁹ showed convincingly that some discontinuities are related to vibrations of the cylinder itself. They suggested that “if there were absolutely no vibration, a Strouhal–Reynolds number plot would have absolutely no discontinuities.”

In the present study we will show that the Strouhal–Reynolds number curve has a single discontinuity that is not caused by *any* of the above suggested mechanisms. Instead,

the discontinuity [see Fig. 1(a) at $Re = 64$] is caused by three-dimensional effects and, in particular, by the phenomenon of oblique vortex shedding, which is controlled to some extent by the end conditions.

The measurements in Fig. 1 have been made with three cylinders (of diameters 0.051, 0.061, and 0.104 cm) in a 6 in. circular test section of an open-jet wind tunnel. The turbulence level was close to 0.1%, with flow uniformity better than 0.3%. Endplate disks of 20:1 diameter ratio were used with cylinder length to diameter ratios up to 240. A good deal of effort was taken to isolate the cylinders from the tunnel, and to damp out any cylinder vibrations. A spectrum analyzer was used to measure frequencies from a hot wire in the wake, and also to display the output of a highly sensitive photodiode setup that monitored cylinder vibrations. (Cylinder vibrations have been measured in this way before.⁹) Oblique-shedding angles were measured using two hot wires, one of which could be traversed the whole span of the wake.

An early indication that flow-induced vibration was not the cause of the discontinuity at $Re = 64$ is deduced by the fact that *all three* cylinders exhibited a discontinuity at the same Re , as shown in Fig. 1(a). Also, the wake spectral measurements were completely unaffected by varying the cylinder natural frequencies at Reynolds numbers both close to and away from the discontinuity. We can conclude from the above that this discontinuity is *not* caused by cylinder vibrations.

Surprisingly, no systematic measurements of oblique shedding angles in the laminar regime have previously been made, although typical angles of 10°–20° have been quoted. The present results in Fig. 1(b) are particularly illuminating, as they show immediately that the Strouhal discontinuity is mirrored by a discontinuous drop in the angle of oblique shedding $|\theta|$. (Note that $|\theta|$ here is measured in the central portion of the cylinder span, and for Re below the discontinuity it corresponds with the frequency f_u .) Above the discontinuity the velocity spectra are “clean” and sharply peaked at the shedding frequency throughout the range $64 < Re < 178$. (Beyond $Re = 178$ there is a three-dimensional transition.) Below $Re = 64$, the spectra exhibit multiple peaks corresponding with “competing” frequencies. One might be tempted, from the form of such spectra, to suggest

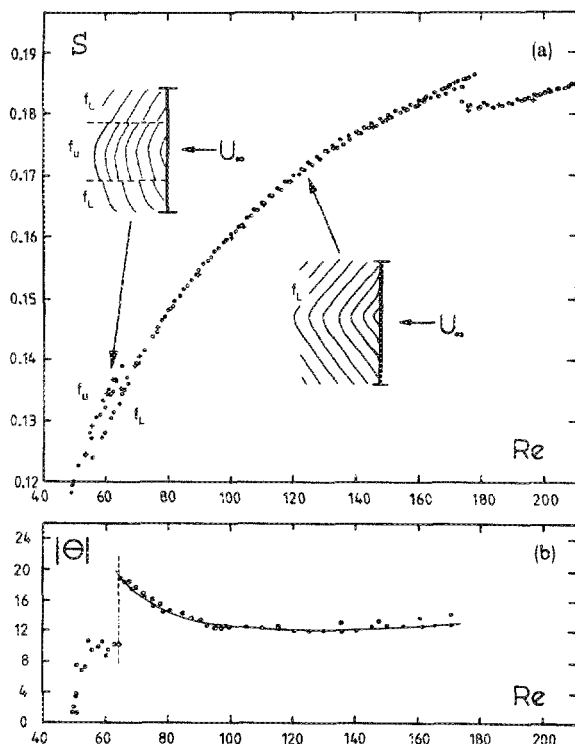


FIG. 1. Strouhal numbers and oblique-shedding angles as a function of Reynolds number. (a) Strouhal numbers (S) versus Reynolds numbers (Re) for three different cylinders: $\circ$, diameter = 0.104 cm ($L/D = 140$); $\bullet$, diameter = 0.061 cm ($L/D = 200$); $+$, diameter = 0.051 cm ($L/D = 240$). (b) Absolute value of oblique-shedding angle $|\theta|$ in degrees versus Re . $\circ$ is for $-\theta$, $\bullet$ is for $+\theta$. Measurements are from the 0.104 cm cylinder. $L/D = 123$, L = cylinder length, D = cylinder diameter.

that cylinder vibrations are present, but in fact there are “cells” of different frequency occurring simultaneously across the span. What happens is that a central cell of frequency f_u (corresponding with the upper Strouhal curve) appears, which is “sandwiched” between two cells of the lower frequency f_L (from the lower Strouhal curve). This corresponds with an overlap between the two Strouhal curves that extends down to $Re \approx 53$, and can be seen in Fig. 1(a). The actual form of the spectra are therefore dependent on the spanwise position. Also, adjacent to each endplate, a further peak in the spectra can be found at a lower frequency, and such a frequency cell has been described by Gerich and Eckelmann.¹⁰ The above results suggest that the cause of this discontinuity is actually a change in the mode of the oblique vortex shedding. The fact that both $+\theta$ and $-\theta$ could be measured at the same Re suggests that the oblique shedding is not, in this case, caused by flow nonuniformities. The oblique shedding is “triggered” by the conditions at the ends of the cylinder, as will be seen below.

In order to support the measurements in the wind tunnel, some further experiments were conducted in the new X-Y Towing Tank at the Graduate Aeronautical Laboratories of the California Institute of Technology. In the tank, flow visualization (using fluorescein dye washed off the surface of cylinders of diameters 0.328 cm and 0.657 cm) showed the changeover of oblique-shedding modes at a value of Re close to 75, slightly higher than in the tunnel. The spanwise cells of different frequency can be seen in Fig. 2(a), with “beat” structures at a low beat frequency ($f_u - f_L$) separating the

central cell of frequency f_u from the cells at frequency f_L to either side. Similar beat structures can be seen at the ends of the cylinder, but at a higher beat frequency of $(f_L - f_e)$ in Fig. 2(b); the phenomenon of “vortex splitting” has been associated with these structures at the ends of the cylinder by Eisenlohr.^{11,12} The inception of the spanwise cell structures caused the Strouhal discontinuity as well as the type of spectra exhibiting multiple peaks. The oblique and steady wake at a single frequency, shown, for example, in Fig. 2(b), caused the “clean” or single-peaked spectra.

Taking into account the above conclusions, one might ask whether a completely continuous Strouhal curve, devoid of any discontinuities, actually exists. One feature that stands out is the fact that the frequency increases as the shedding angle decreases, a trend that was also clear in the work of Ramberg.¹³ As a first step we can transform the Strouhal data (S_θ) of Fig. 1(a), using the data for the oblique angles (θ) given in Fig. 1(b), by the transformation

$$S_0 = S_\theta / \cos \theta. \quad (1)$$

Actually, this simple formula is supported by linear stability analysis, and its use here was suggested by Brown.¹⁴ (Further work concerning an explanation of the oblique

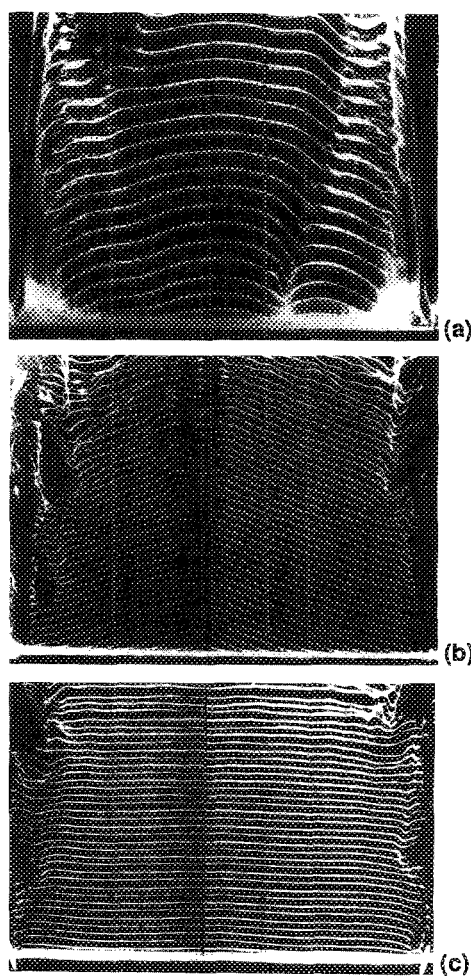


FIG. 2. Visualization of the different modes of laminar vortex shedding. Flow is from bottom to top. (a) Oblique-shedding mode at $Re = 60$ (below the discontinuity). (b) Oblique-shedding mode at $Re = 85$ (above the discontinuity). (c) Parallel-shedding mode at $Re = 110$. This is “induced” to occur by angling the endplates (a) $L/D = 70$; (b) $L/D = 140$; (c) $L/D = 130$.

shedding will appear in a forthcoming paper.) The transformed Strouhal curve for S_0 in Fig. 3 (the open circles) is now close to being continuous (smoothing out the previous discontinuity at $Re = 64$). If this new curve is "universal," then it should be possible to transform other Strouhal data onto this curve by using corresponding angles (θ) and Eq. (1). The second step in showing the existence of such a continuous universal curve was to induce parallel shedding (i.e., $\theta = 0$) in the wind tunnel, over the entire range of laminar Re , and then show that the ensuing Strouhal numbers lie directly on the curve for S_0 . It was found, before this point in the study, that spans up to 240 diam were influenced by the end conditions. By angling-in the front end of the 10:1 end-plates by around 12° – 15° , the vorticity at the ends was stretched axially to cause an end boundary condition of parallel shedding; this then "induced" parallel shedding over the whole span. This dramatic change to parallel shedding is shown clearly in Fig. 2(c), which can be contrasted with the oblique shedding of Fig. 2(b). (Recently Eisenlohr¹² has also induced parallel shedding by ending his cylinder span with larger-diameter cylinders, and found higher Strouhal numbers than for oblique shedding, a trend which agrees with the results below.

The present curve of Strouhal numbers corresponding to the parallel shedding (the solid circles in Fig. 3) showed remarkably good agreement with the transformed oblique-shedding data (a result that suggests that the Strouhal curve is indeed universal for a circular cylinder). A further feature of the parallel shedding is that at every point of the universal curve the spectra are "clean" and the curve is correspondingly smooth (i.e., without any discontinuities) over the whole Re range. The smoothness of the Strouhal variation was checked several times with several different tensions in the cylinder. [Also, as an aside comment, we can see that it is not correct to assume that the upper curve (f_u) near the discontinuity in Fig. 1(a) corresponds with the existence of parallel shedding over the whole span.]

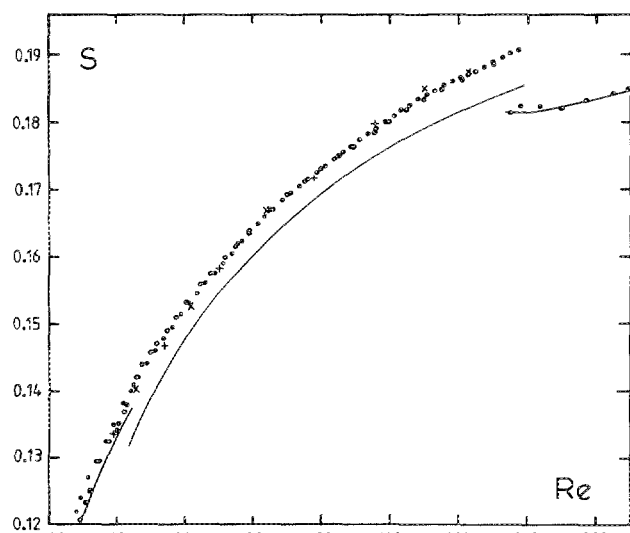


FIG. 3. The "universal" Strouhal curve for a circular cylinder. Measurements from the wind tunnel: $\bullet$, parallel-shedding data (diameter = 0.104 cm, $L/D = 90$); $\circ$, transformed oblique-shedding data using the formula $S_0 = S_\theta / \cos \theta$. Data are from the 0.104 cm cylinder, $L/D = 123$; —, curves through the original S_θ data. Measurements from the X-Y Towing Tank: $\times$, 0.657 cm cylinder, $L/D = 72$; $+$, 0.328 cm cylinder, $L/D = 140$.

As a final check, and the third step in showing the existence of a universal curve, it was decided to undertake frequency measurements for parallel shedding in a distinctly different facility. The X-Y Towing Tank was then used with the 0.328 cm and 0.657 cm cylinders, and with a video system that has an incorporated timer unit. The results (shown as crosses in Fig. 3) agree well with the other data, although the frequencies could not be measured to the same accuracy as in the wind tunnel.

Another way to determine the universal curve (i.e., parallel shedding) is from two-dimensional numerical calculations. Such calculations have been made by Karniadakis and Triantafyllou.¹⁵ Their Strouhal curve lies 5%–15% above the universal curve here, and this disparity is not understood. (They do not give an error analysis for their results; the errors in determining the present curve are best indicated by the small scatter of the plotted points in Fig. 3.) For a comparison with the numerical computations, the Strouhal-Reynolds number relation for the universal curve is given closely by $S = A/Re + B + CRe$, where $A = -3.3265$, $B = 0.1816$, and $C = 1.600 \times 10^{-4}$.

It is concluded that if very careful attention is paid to damping any cylinder vibrations, and that if the free stream is sufficiently uniform, then the Strouhal curve for parallel shedding should be completely smooth and devoid of discontinuities. If the end conditions are such that the shedding is oblique, then a change in the mode of oblique shedding causes a discontinuity to occur. If, however, these data are transformed using Eq. (1), then they fall onto the smooth parallel-shedding curve. The close agreement between the transformed oblique-shedding data with the parallel-shedding data in the tunnel, and also with the parallel-shedding data from the towing tank, provides strong evidence that the Strouhal curve (S_0) is universal as well as continuous.

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¹A. Roshko, NACA Report 1191, 1954.

²D. J. Tritton, *J. Fluid Mech.* **6**, 547 (1959).

³E. Berger, *Z. Flugwiss.* **12**, 41 (1964).

⁴M. Gaster, *J. Fluid Mech.* **38**, 565 (1969).

⁵M. Gaster, *J. Fluid Mech.* **46**, 749 (1971).

⁶C. Mathis, M. Provansal, and L. Boyer, in *Proceedings of the Second International Symposium on Applications of Laser Anemometry to Fluid Mechanics* (Ladoan Instituto Superior Tecnico, Lisbon, 1984), p. 12.5.

⁷J. H. Gerrard, *Philos. Trans. R. Soc. London Ser. A* **288**, 351 (1978).

⁸K. R. Sreenivasan, in *Frontiers in Fluid Mechanics*, edited by S. H. Davis and J. L. Lumley (Springer, Berlin, 1985), p. 41.

⁹C. W. Van Atta and M. Gharib, *J. Fluid Mech.* **174**, 113 (1987).

¹⁰D. Gerich and H. Eckelmann, *J. Fluid Mech.* **122**, 109 (1982).

¹¹H. Eisenlohr, Diplom. thesis, Max-Planck-Institut für Strömungsforschung, Göttingen, 1986.

¹²H. Eisenlohr (private communication).

¹³S. E. Ramberg, *J. Fluid Mech.* **128**, 81 (1983).

¹⁴G. L. Brown (private communication).

¹⁵G. E. Karniadakis and G. S. Triantafyllou, submitted to *J. Fluid Mech.*